Definition:

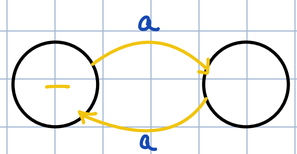
↳ Finite Automata (FA) is a collection of three things:

1. Finite set of states
   - ↳ Initial state → only one
   - ↳ Final state → Can be many
2. An alphabet of possible input letters.
3. Finite set of transitions that tell for each letter of the input alphabet which state to go next.

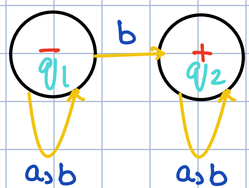
How it works:

1. Take an input of string of letters
2. Reads from left to Right
3. Beginning letter at the start state, the letters determine the sequence of states.
4. Sequence ends when last input is read.

↳ A string that leaves us in the final state is called the language defined by the finite automation.



↳ This Automata doesn't take us to the final state. So, strings 'aa' are not accepted by this FA.



↳ This automata does take us to the final state. So string 'abba' is accepted by this FA.

Incoming transitions for each state

$(a+b)^* b (a+b)^* \rightarrow$

$$q_1 = \Lambda + q_1 a + q_1 b \quad \text{—— } (1)$$

$$q_2 = q_1 b + q_2 a + q_2 b \quad \text{—— } (2)$$

$$q_1 = \overset{Q}{\Lambda} + \overset{R}{q_1}\overset{P}{(a+b)} \qquad QP^*$$

$$(a+b)^*$$

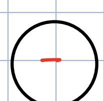
$$q_2 = \overset{Q}{(a+b)^* b} + \overset{R}{q_2}\overset{P}{(a+b)}$$

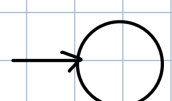
$$= (a+b)^* b (a+b)^*$$

Solve for final

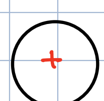
Notation:

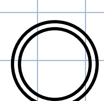
1. Initial state:





2. Final state:





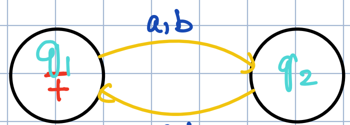
state

Ways to study FA:

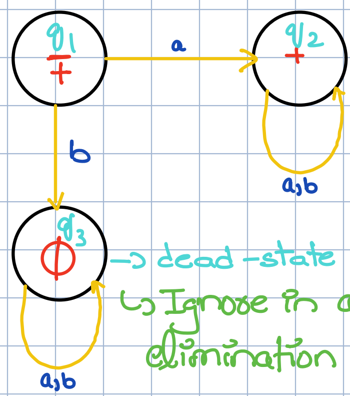
- Given a Language, can we build a machine FA.
- Given a machine, can we deduce it's language.

Example,

① $L = \{$All words of even length$\}$



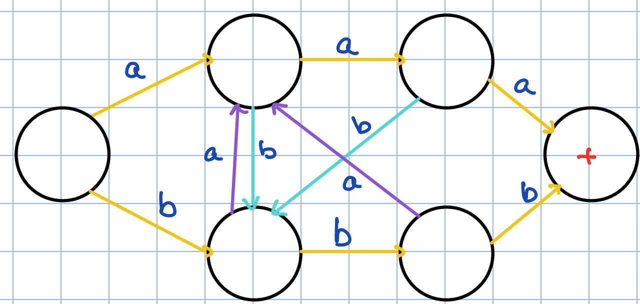
② $a(a+b)^*$



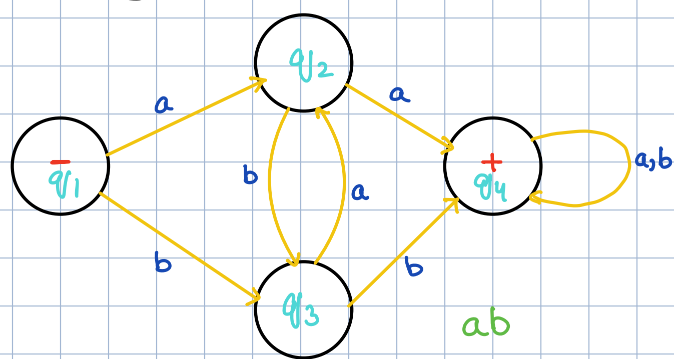
→ dead-state

↳ Ignore in above elimination method

③ $L = \{$ All words that contain 'aaa' or 'bbb'$\}$



④ Determine what language is accepted by the following FA.



ab

*

$aa(a+b)^* + bb(a+b)^* + abb(a+b)^* + baa(a+b)^*$

$(a+b)^*(aa+bb)(a+b)^*$

$q_1 = \Lambda$

$q_2 = q_1a + q_3a$

$q_3 = q_1b + q_2b$

$q_4 = q_2a + q_3b + q_4a + q_4b$

$q_4 = aa^*a + bb^*b + q_4(a+b)$

$= aa^*a + bb^*b + (a+b)^*$

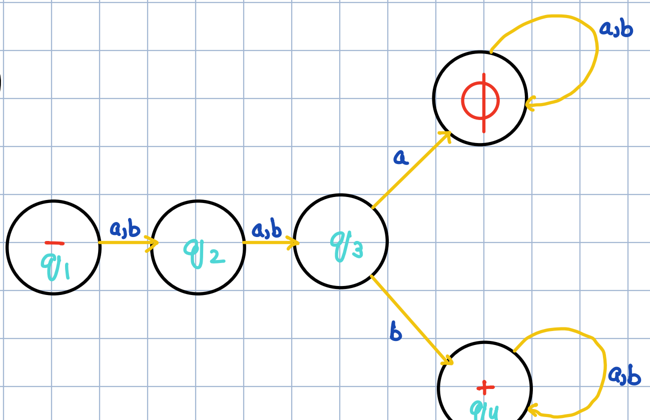
$q_2 = a + q_3a$

$= aa^*$

$q_3 = bb^*$

∴ Must have double letters

⑤



$(a+b)(a+b)b(a+b)^*$

$q_1 = \Lambda$

$q_2 = q_1a + q_1b$

$= q_1(a+b)$

$q_3 = q_2a + q_2b$

$= q_2(a+b)$

$= q_1(a+b)(a+b)$

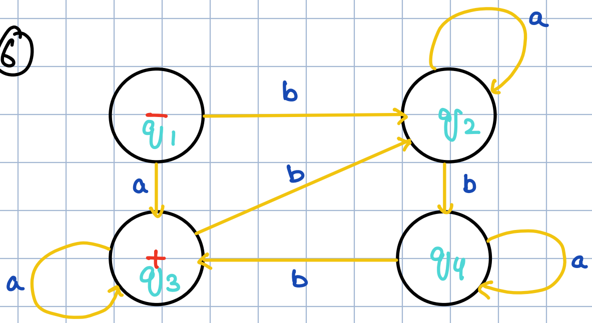
$= (a+b)(a+b)$

$q_4 = q_3b + q_4a + q_4b$

$= (a+b)(a+b)b + q_4(a+b)$

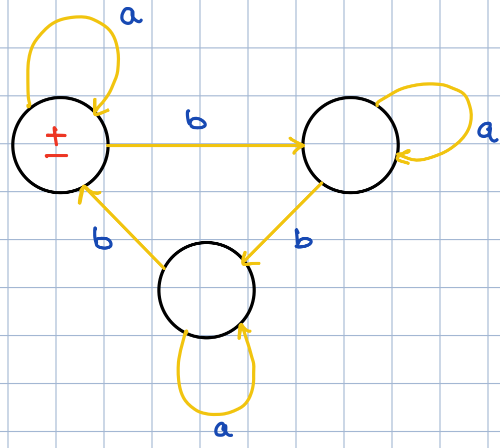
$= (a+b)(a+b)\,b\,(a+b)^*$

(6)



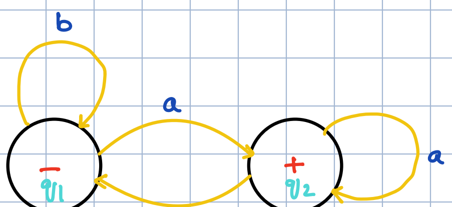
$ba^*ba^*ba^* + aa^*[ba^*ba^*ba^*]^*$

(7)



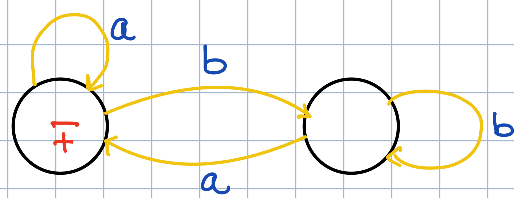
$(a^* + ba^*ba^*b)^*$

(8)



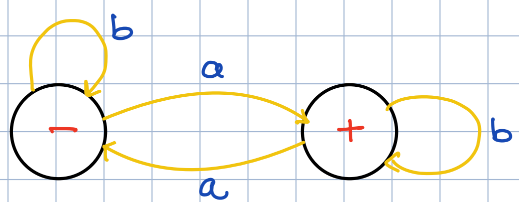
$b^*aa^*(bb^*aa^*b)^*$

(9)



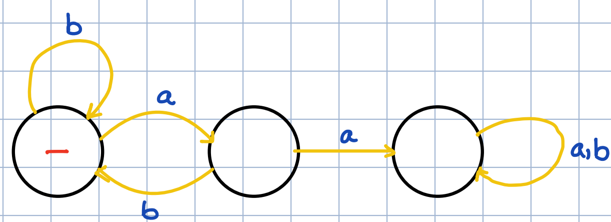
$(a^* + bb^*a)^*$

(10)



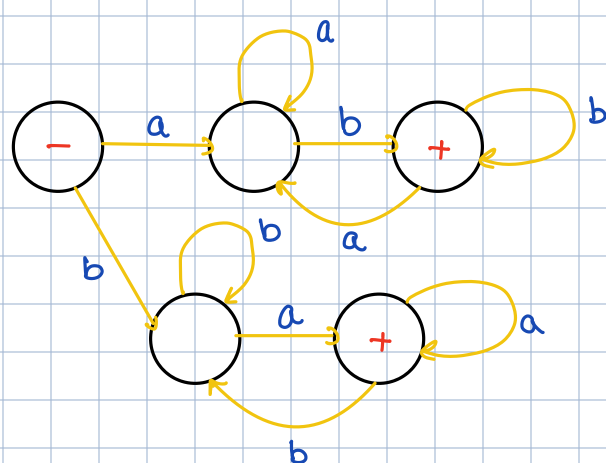
$b^*ab^*(ab^*ab^*)^*$

(11)

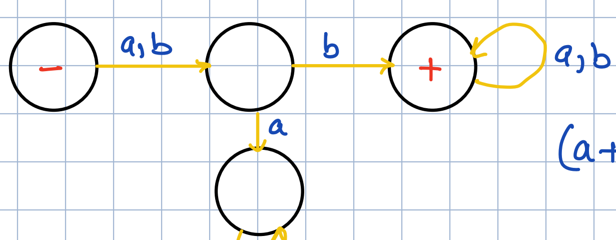


$b^*a(bb^*a)^*a(a+b)^*$

(12)

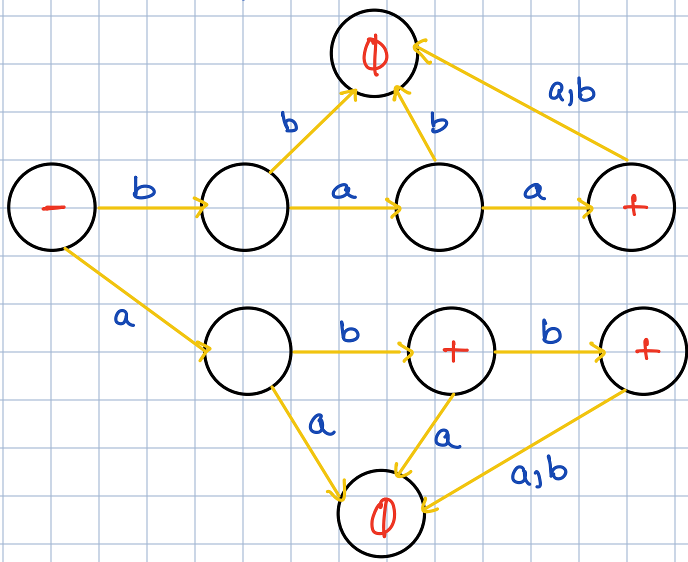


Q2.



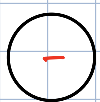
$(a+b)b(a+b)^*$

Q3.

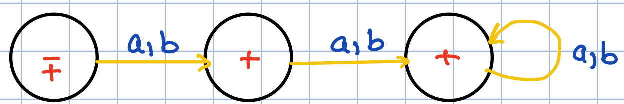


Q4.

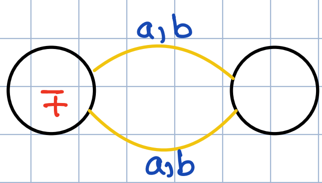
(i)



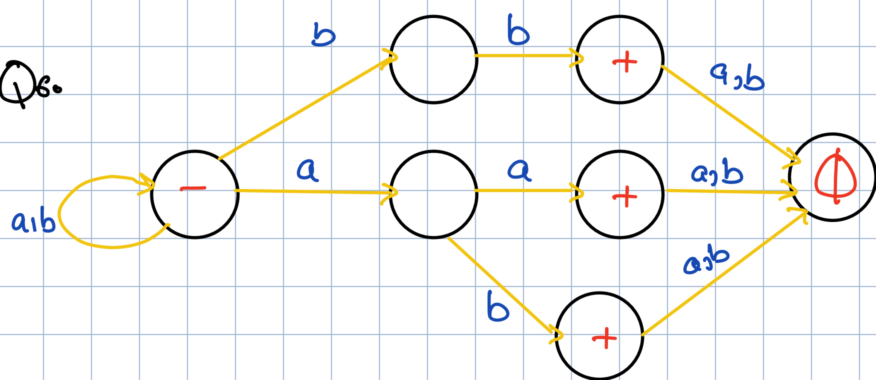
(ii)



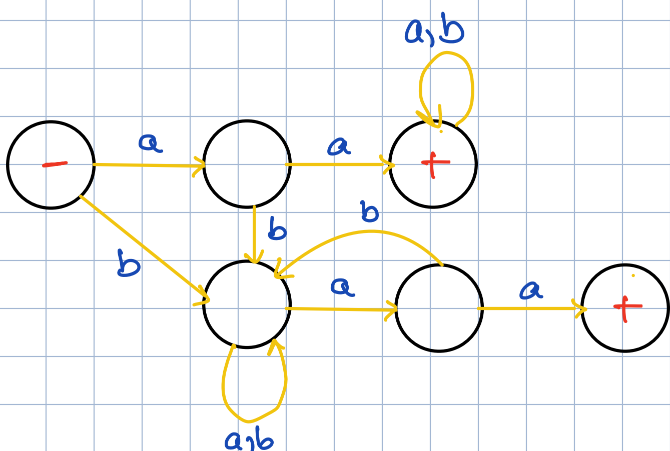
Q5.



Q6.

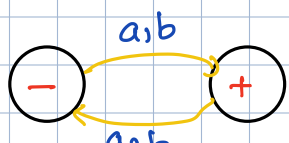


Q7.

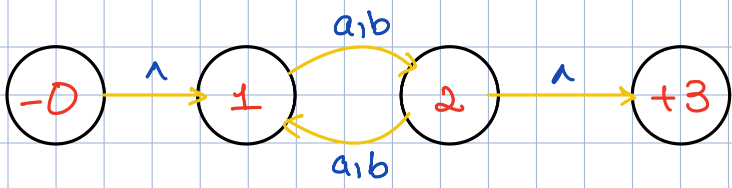


Q18.

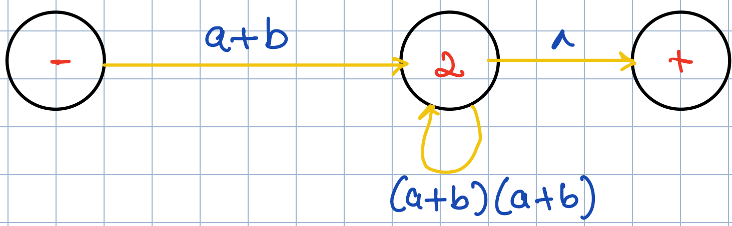
(i)



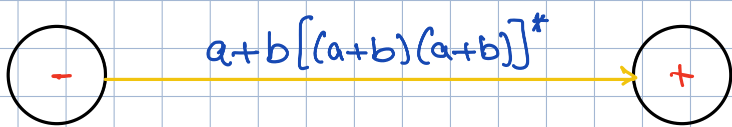
$(a+b)^*$



-0
∧
1
a,b
a,b
2
∧
+3

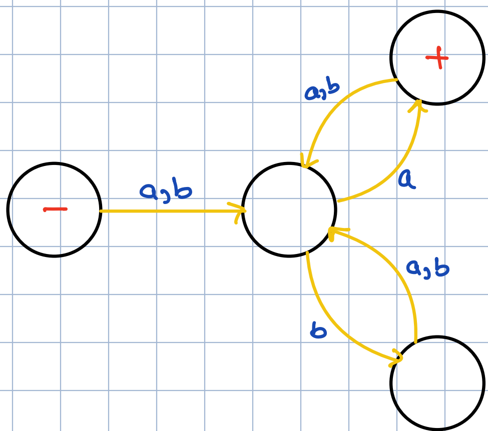


-
a+b
2
∧
+
(a+b)(a+b)

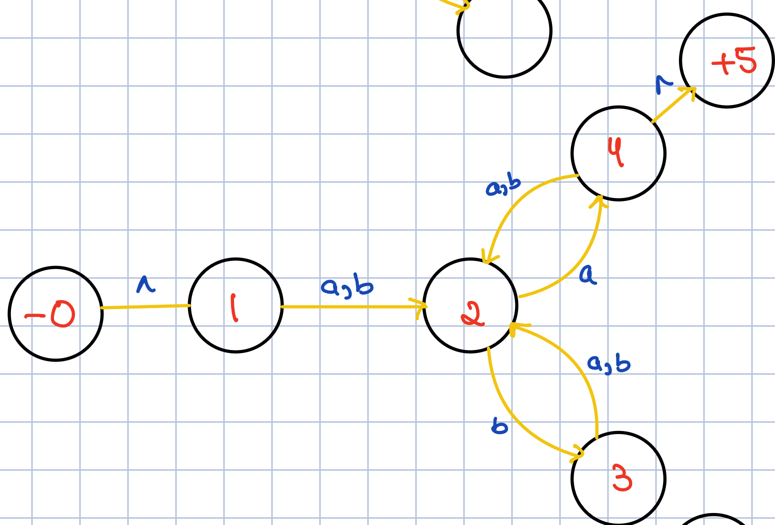


-
a+b[(a+b)(a+b)]*
+

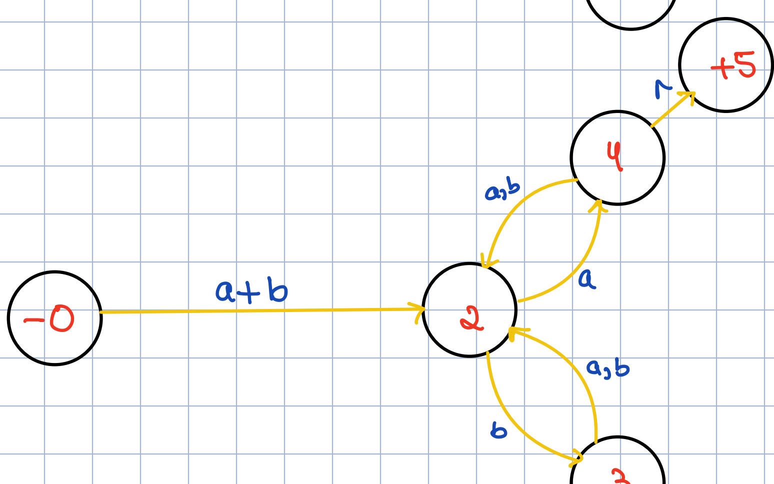
(ii)



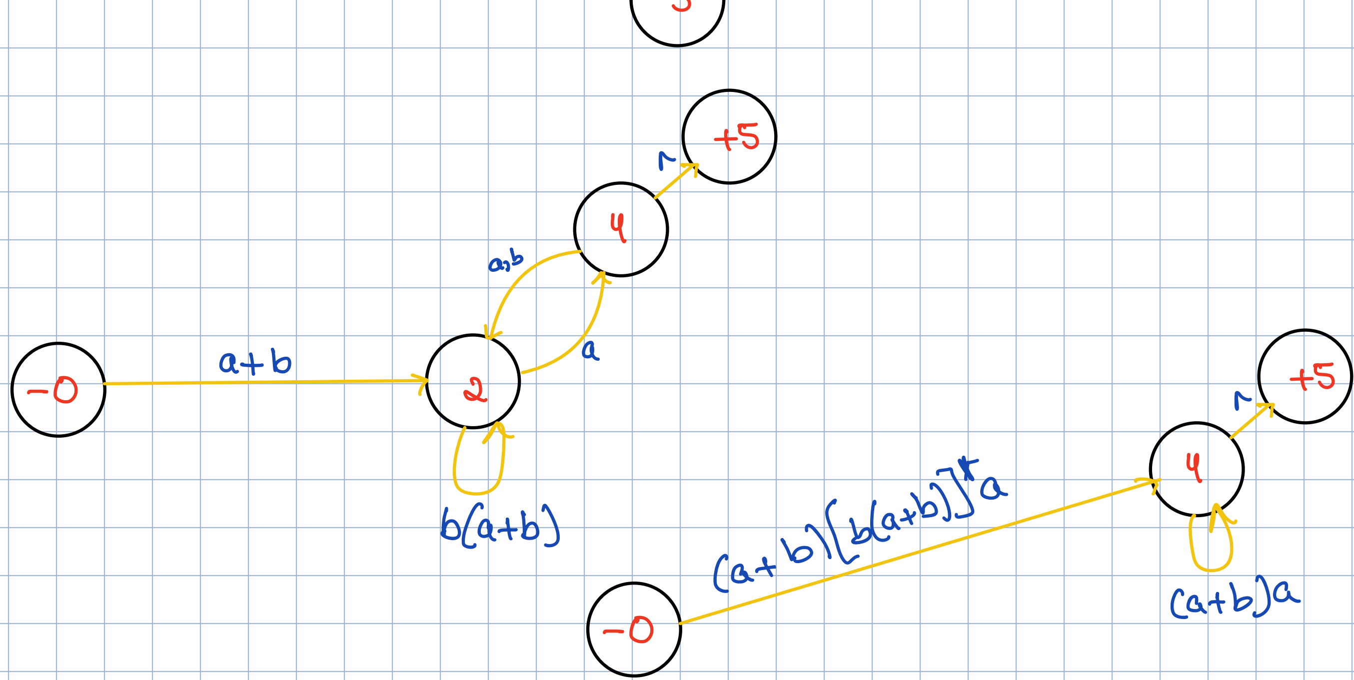
+
a,b
-
a,b
a
a,b
b



+5
∧
4
a,b
-0
∧
1
a,b
a
2
a,b
b
3

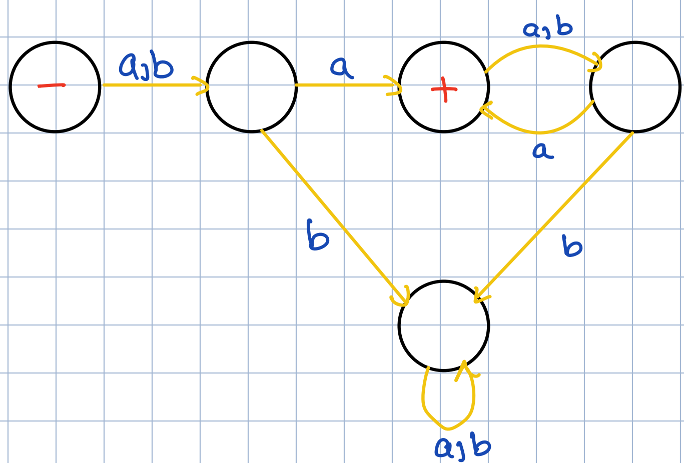


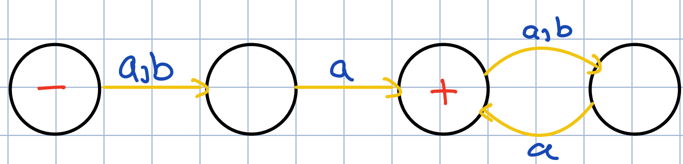
+5
∧
4
a,b
a
-0
a+b
2
a,b
b
3

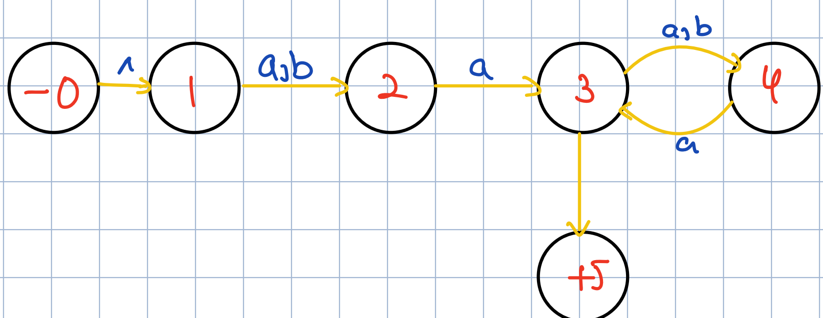


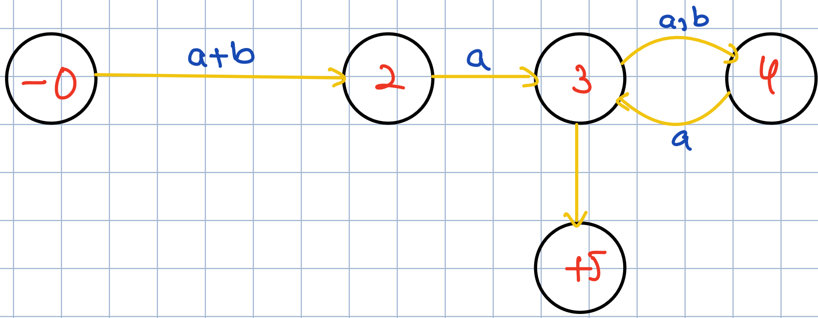
$$(a+b)\left[b(a+b)\right]^{*}a\left[(a+b)a\right]^{*}$$

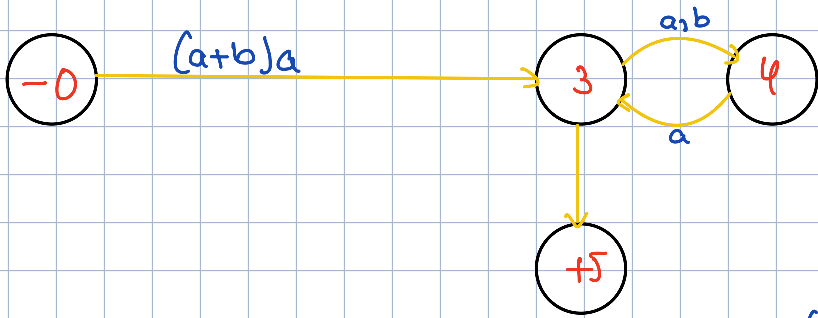
iii)

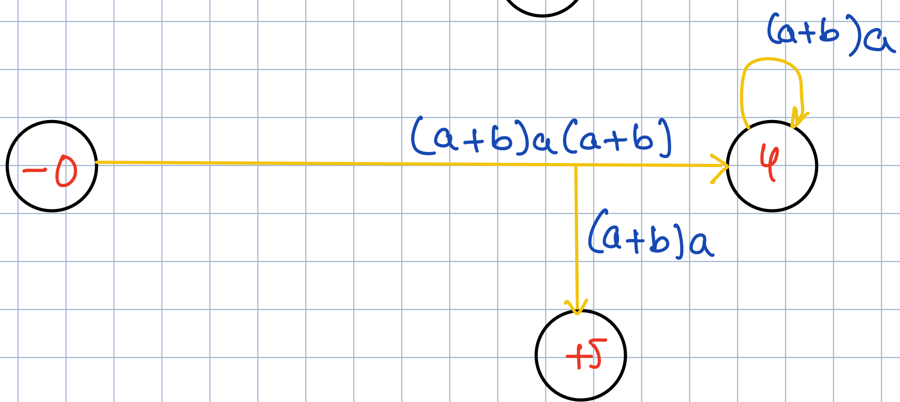












$$(a+b)a + (a+b)a(a+b)[a(a+b)]^*$$

$$(a+b)a[(a+b)a]^*$$